Reducing Employee Attrition at Globex Pharma: A Predictive Analytics Approach using Design Thinking

(DPBS1190 Assessment 3: Group Project Report)

Academic Integrity Declaration

"I declare that the work I have submitted relating to this assessment task is completely of my own and I confirm that I have complied with all requirements of UNSW Academic Honesty and plagiarism policy".

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EXECUTIVE SUMMARY

This project investigates employee attrition at Globex Pharma using data-driven predictive models to support strategic HR decision making. We applied logistic regression model and classification tree analysis to identify key predictors of employee attrition and uncover vulnerable employee segments. This allows us to gain insights on attrition issue both qualitatively and quantitatively.

Our finding reveals that Overtime, Job Involvement, Environment satisfaction and Years Since Last Promotion are the most significant variables that influencing attrition. These factors were selected based on both statistical significance and their real -world practical relevance in employee attrition.

Three vulnerable groups were identified with each combining promotion delays with moderate to low job involvement and working overtime. This also further highlight the risk of career stagnation, burnout and misalignment of rewards and engagement.

Based on these findings, we provide recommendations focused on reducing attrition by addressing workload balance and low job involvement issue.

Our approach combined with design and agile thinking by framing the problem from a human-centered perspective, testing and iterating models and adapting our approach based on Globex Pharma's real-world business needs. Ethical considerations concerning data privacy.....

Eventually, this project shows how data-driven insights can empower better workforce management and enables Globex Pharma to address employee attrition effectively.

INTRODUCTION

Big data enables businesses to transition from guessing to knowing when making decisions. Instead of relying only on experience or intuition, managers can now use patterns in large datasets to spot risk early and take initiative. At Globex Pharma, rising employee attrition has raised concerns about staff satisfaction and company long-term operation stability and efficiency. By applying predictive analysis to employee data, Globex Pharma can better identify what is driving the attrition. This approach enables more targeted and timely decisions which helps HR teams focus their efforts on where it matters the most before the issues become costly. (Wamba et al., 2015).

BUSINESS ISSUE & PROJECT GOAL

Project Goal

To develop predictive models for likelihood of employees' attrition. Identify key variables in determining staff turnover Globex Pharma, with the aim of providing actionable insights to address employee dissatisfaction whilst implementing strategies to improve retention.

Main questions

1.Which variables are strongly associated with employee attrition?

2.How well do our predictive models perform in identifying potential attrition cases?

3. Which employee groups are most vulnerable to attrition and what actionable recommendations can we provide?

DATA ANALYSIS 1 – VARIABLE SELECTION APPROACH

We used logistics regression to identify key variables that significantly affects employee attrition within the dataset.

MODEL 1: ALL VARIABLES

```
Call:
glm(formula = Attrition ~ ., family = "binomial", data = train)

Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                     -1.218e+01  7.989e+02  -0.015 0.987833
Age                             -3.373e-02  1.821e-02  -1.852 0.064041 .
BusinessTravelTravel_Frequently  2.440e+00  5.674e-01   4.301 1.70e-05 ***
BusinessTravelTravel_Rarely      1.516e+00  5.196e-01   2.918 0.003525 **
DepartmentResearch & Development 1.391e+01  7.989e+02   0.017 0.986110
DepartmentSales                  1.403e+01  7.989e+02   0.018 0.985985
DistanceFromHome                 4.931e-02  1.447e-02   3.408 0.000655 ***
Education                        4.784e-03  1.152e-01   0.042 0.966864
EnvironmentSatisfaction         -4.738e-01  1.097e-01  -4.321 1.56e-05 ***
GenderMale                       1.898e-01  2.473e-01   0.767 0.442837
JobInvolvement                  -6.349e-01  1.698e-01  -3.739 0.000185 ***
JobRoleHuman Resources           1.598e+01  7.989e+02   0.020 0.984041
JobRoleLaboratory Technician     1.999e+00  6.233e-01   3.207 0.001343 **
JobRoleManager                   2.822e-02  1.402e+00   0.020 0.983936
JobRoleManufacturing Director    2.988e-01  6.896e-01   0.433 0.664795
JobRoleResearch Director        -9.545e-01  1.199e+00  -0.796 0.426048
JobRoleResearch Scientist        1.157e+00  6.248e-01   1.851 0.064142 .
JobRoleSales Executive           1.230e+00  1.724e+00   0.713 0.475723
JobRoleSales Representative      2.383e+00  1.765e+00   1.350 0.176911
MaritalStatusMarried            -4.628e-02  3.396e-01  -0.136 0.891596
MaritalStatusSingle              1.122e+00  4.344e-01   2.584 0.009764 **
MonthlyIncome                    8.086e-05  8.287e-05   0.976 0.329179
NumCompaniesWorked               1.888e-01  5.148e-02   3.668 0.000245 ***
OverTimeYes                      1.872e+00  2.541e-01   7.365 1.77e-13 ***
PerformanceRating               -3.777e-01  3.507e-01  -1.077 0.281474
RelationshipSatisfaction        -2.892e-01  1.093e-01  -2.646 0.008149 **
StockOptionLevel                -1.564e-01  2.015e-01  -0.776 0.437792
TotalWorkingYears               -6.262e-02  3.708e-02  -1.689 0.091292 .
TrainingTimesLastYear           -2.078e-01  9.379e-02  -2.216 0.026690 *
WorkLifeBalance                 -5.667e-01  1.710e-01  -3.314 0.000919 ***
YearsAtCompany                   7.200e-02  5.397e-02   1.334 0.182202
YearsInCurrentRole              -1.499e-01  6.494e-02  -2.308 0.020980 *
YearsSinceLastPromotion          2.444e-01  5.901e-02   4.141 3.46e-05 ***
YearsWithCurrManager            -1.513e-01  6.254e-02  -2.418 0.015589 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Model 1: All variables

We began by fitting a model with all available predictors. From this, we examined the **p-values** to identify significant contributors to attrition.

Variables	P-value
Over Time	1.77e-13
Environment Satisfaction	1.56e-05
Employee who travels frequently for business	1.70e-05
Year Since Last Promotion	3.46e-05
Job Involvement	0.000185
Number of companies worked	0.000245
Distance from home	0.000655
Work Life Balance	0.000919

Table 1: Significant predictors (***) included (from high to low significance)

We then narrowed down the model to a more interpretable and business-relevant set of 4 predictors:

Selected Variables	Reason for Inclusion	Relation to Attrition
OverTime	High P significance; indicates workload pressure.	Burnout, poor work life balance.
Environment Satisfaction	Engagement with work environment.	Negative or dissatisfying work environment cause employee to seek employment elsewhere.
JobInvolvement	Reflects how engaged employees feel with their work.	Low involvement may lead to feeling disconnected and demotivated.
YearsSinceLastPromotion	Indicates career stagnation.	Lack of recognition cause employees feel undervalued or unappreciated eventually leave for better role.

Table 2: Variable Selection

Our chosen variables align closely with identified variables in HR research (Ruehl 2024).

Justification:
We excluded other significant variables such as **BusinessTravel, Number_of_companies_worked, Distance_from_home** as they are either job role-dependent or employee-specific factor outside company influence. **Work_Life_Balance** was omitted due to conceptual overlap with OverTime. Overall, these factors offer limited scope for actionable intervention or managerial control by Globex despite their high significant level.

DATA ANALYSIS 2 - LOGISTIC REGRESSION MODEL

Use our selected 4 variables to build a logistic regression model.

```
> summary(model1)

Call:
glm(formula = Attrition ~ OverTime + YearsSinceLastPromotion +
    EnvironmentSatisfaction + JobInvolvement, family = "binomial",
    data = train)

Coefficients:
                        Estimate Std. Error z value Pr(>|z|)
(Intercept)             -0.02927    0.44355  -0.066 0.947379
OverTimeYes              1.49072    0.20232   7.368 1.73e-13 ***
YearsSinceLastPromotion  0.01121    0.03057   0.367 0.713724
EnvironmentSatisfaction -0.33490    0.08906  -3.760 0.000170 ***
JobInvolvement          -0.47263    0.13729  -3.443 0.000576 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 729.42  on 799  degrees of freedom
Residual deviance: 652.43  on 795  degrees of freedom
AIC: 662.43

Number of Fisher Scoring iterations: 5
```

Model 2: Logistic Regression Model

*We predicted attrition probability on test data and chose a threshold of **0.16**, matching the base attrition rate (16%) in the dataset.*

```
> confusionMatrix(factor(class1, levels = c("Yes","No")),test$Attrition, positive="Yes"
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  116  11
       Yes  56  17

               Accuracy : 0.665
                 95% CI : (0.595, 0.73)
    No Information Rate : 0.86
    P-Value [Acc > NIR] : 1

                  Kappa : 0.1683

 Mcnemar's Test P-Value : 7.639e-08

            Sensitivity : 0.6071
            Specificity : 0.6744
         Pos Pred Value : 0.2329
         Neg Pred Value : 0.9134
             Prevalence : 0.1400
         Detection Rate : 0.0850
   Detection Prevalence : 0.3650
      Balanced Accuracy : 0.6408

       'Positive' Class : Yes
```

Confusion Matrix 1: Model Evaluation – Logistic Regression

Explanation

The logistic regression model predicts employees attrition based on Overtime, Job Involvement, Environment Satisfaction and Years Since Last Promotion.

Model Summary

The logistic regression model summary provides following key insights:

Key Terms from the Matrix:	
True Positive (TP) = 17	**Predicted to leave** and **actually left**
False Positive (FP) = 56	**Predicted to leave** but **actually stayed**
False Negative (FN) = 11	**Predicted to stay** but **actually left**
True Negative (TN) = 116	**Predicted to stay and actually stayed**

1. **Coefficients and Significant:**
   - **Intercept:** Not significant (p value = 0.947)
   - **Overtime:** p-value=1.73e-13<0.05; Positive coefficient(1.49) -> statistically significant+positive correlation
   - **Years Since Last Promotion:** p-value=0.714; Positive coefficient(0.1121) -> not statistically significant+positive correlation

- **Environment Satisfaction:** p-value=-0.3349<0.05; Positive coefficient(0.000170) -> statistically significant+positive correlation
- **Job Involvement:** p-value=0.000576<0.05; Negative moderate correlation(-0.47263) -> statistically significant+negative correlation

2. **Model Performance:**
    - **Accuracy:0.665**, indicating that the model correctly predicts employee attrition about 66.5% of the time.
    - **Sensitivity:0.6071**, indicating that the model correctly identifies about 60.7% of employee attrition (employee left).
    - **Specificity:0.6744**, indicating that the model correctly identifies about 67.4% of employees stayed.
    - **Balanced Accuracy:64.1%,** which is the average of sensitivity and specificity.

Strategic Insights for Reducing Attrition

1. **Manage Overtime**
   Employees working overtime are significantly more likely to leave. Consider reviewing workload and overtime policies.

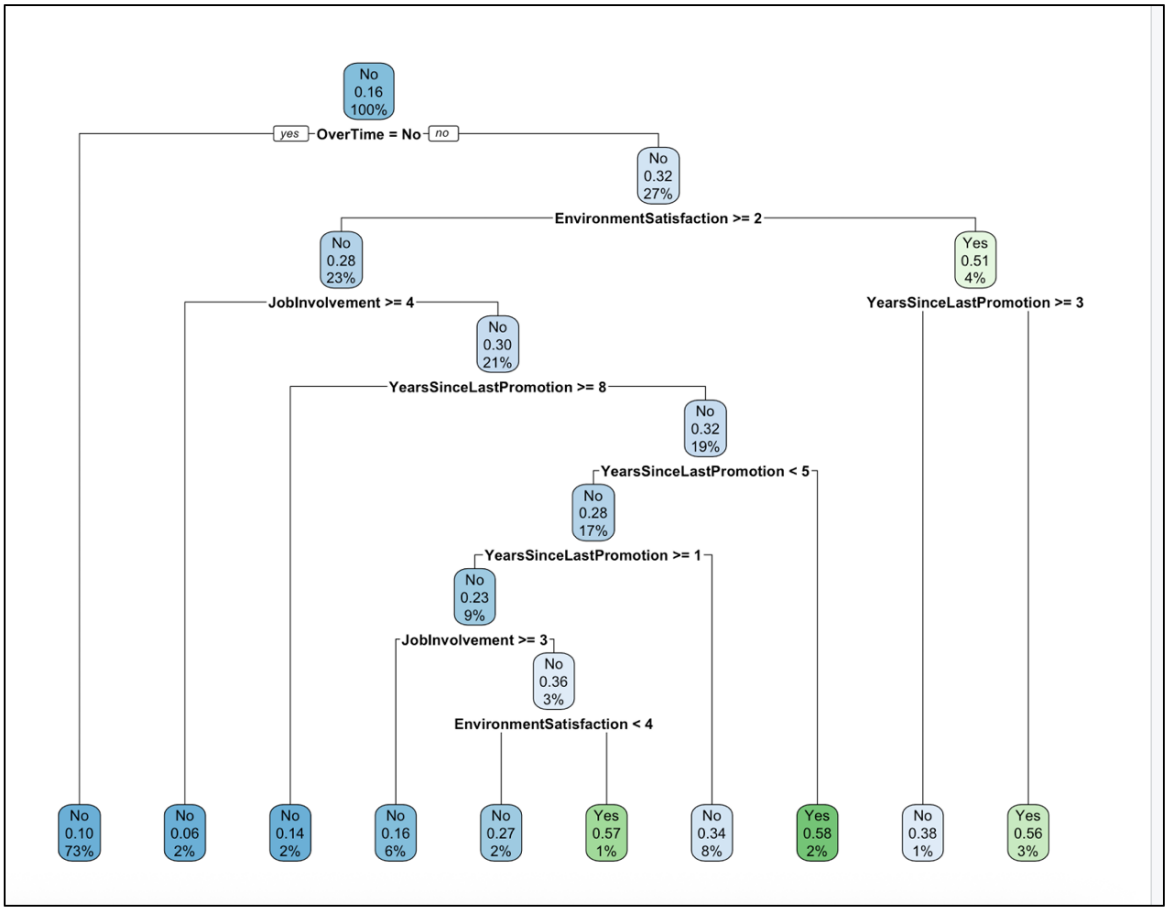
2. **Promotion Gap (not statistically significant)**
   Although not statistically significant, years since last promotion showed a small positive effect on attrition (0.1121). Consider offering **career growth opportunities** or development programs.

3. **Enhance Work Environment**
   Environment Satisfaction is statistically significant. Continue to invest in workplace conditions, wellbeing, and team support.

4. **Boost Job Involvement**
   Low job involvement strongly predicts attrition (negative correlation). Engage employees through meaningful work, role clarity, and skills-development training.

5. **Model Limitations**
   - While accuracy is **fair but acceptable (66.5%)**, the **low precision (23.3%)** indicates the model generates **many false positives**. This means that out of all employees predicted to leave, **only 23.3% actually did.**
   - Sensitivity (60.7%) is lower than specificity (67.4%), meaning the model is **better at identifying who will stay than who will leave.**
   - It is useful for identifying trends but **should not be the sole tool** for employee retention decisions.

DATA ANALYSIS 3 – CLASSIFICATION TREE



Model 3: Classification Tree

Data Preparation

1. **Factor Conversion**: The Attrition variable was treated as a factor to allow classification modelling.
2. **Train-Test Split**: The data was split into training (80%) and testing (20%) sets using a random seed to ensure reproducibility.

Decision Tree Model Summary

- **Model**: A classification tree was built using the rpart package to predict **Attrition** based on same four variables:
  OverTime, JobInvolvement, EnvironmentSatisfaction, and YearsSinceLastPromotion.
- **Control Parameter**: The complexity parameter (cp) was set to **0.001** to allow the tree to grow deeper and capture more splits.

- **Variable Importance**:

```
> summary(tree_model)
Call:
rpart(formula = Attrition ~ OverTime + EnvironmentSatisfaction +
    JobInvolvement + YearsSinceLastPromotion, data = train_data,
    method = "class", control = rpart.control(cp = 0.001))
  n= 800

           CP nsplit rel error   xerror       xstd
1 0.007692308      0 1.0000000 1.000000 0.08026399
2 0.002564103      6 0.9538462 1.115385 0.08381395
3 0.001000000      9 0.9461538 1.115385 0.08381395

Variable importance
               OverTime YearsSinceLastPromotion EnvironmentSatisfaction          JobInvolvement
                     54                      19                      16                      11
```

  - **OverTime**:54%
  - **YearsSinceLastPromotion**:19%
  - **EnvironmentSatisfaction**:16%
  - **JobInvolvement**:11%

- **Tree Structure**:

  - The **root node** splits on OverTime, showing it is the most influential variable.
  - Further splits are based on EnvironmentSatisfaction, JobInvolvement, and YearsSinceLastPromotion.
  - Most terminal nodes (leaves) predict No (employee stays), reflecting the dataset's class imbalance.

Model Evaluation-Confusion Matrix and Statistics

```
> print(conf_matrix)
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  162  28
       Yes   4   6

               Accuracy : 0.84
                 95% CI : (0.7817, 0.8879)
    No Information Rate : 0.83
    P-Value [Acc > NIR] : 0.3963

                  Kappa : 0.2118

 Mcnemar's Test P-Value : 4.785e-05

            Sensitivity : 0.1765
            Specificity : 0.9759
         Pos Pred Value : 0.6000
         Neg Pred Value : 0.8526
             Prevalence : 0.1700
         Detection Rate : 0.0300
   Detection Prevalence : 0.0500
      Balanced Accuracy : 0.5762

       'Positive' Class : Yes
```

Confusion Matrix 2: Model Evaluation – Classification Tree

Insights

1. The model achieved a good overall **accuracy of 84%**, correctly predicting most employees who stayed.
2. The model shows **high specificity** (97.59%) but **low sensitivity** (17.65%), meaning it is much better at predicting employees who stay than those who leave.
3. **OverTime** is the most important predictor, suggesting employees required to work overtime are more likely to leave.

Vulnerable Groups

Our classification tree identified three key groups with elevated attrition risk. (From right to left of the tree)

Group A: OverTime, EnvironmentSatisfaction<2, YearsSinceLastPromotion<3

Group A (3% of dataset, Predicted chance of attrition: 56%)

Employees who work overtime, have very low environment satisfaction (<2), and were promoted within the past 3 years. Despite recent promotions, these group of employees are still likely to leave due to increased responsibilities and a poor work environment leading to increase in burnout. This indicates that promotions alone may not be sufficient to offset poor working conditions or excessive workload. (Wilson 2023)

Group B: OverTime, EnvironmentStatisfaction>=2, Job Involvement<4, YearsSinceLastPromotion between 5 to 8 years

Group B (2% of dataset, Predicted chance of attrition: 58%)

Employees who worked overtime, promoted between 5-8 years, environment satisfaction of more than 2 and reported only moderate job involvement (< 4).
This suggests a group that possibly has career stagnation. They have waited several years without promotion while continuing to take on high workloads, leaving them increasingly frustrated and unsatisfied. This group of people also has moderate job involvement, which indicates they are less emotionally connected to their roles. This could be due to the work becoming repetitive, insufficiently stimulating. When long hours and efforts are not aligned with career growth, employees may begin to feel pressurized (Ono.2023), and their works are repetitive. Over time, this degradation of engagement can lead to attrition in Globex.

Another key observation from the classification tree is that employees who have gone 8+ years without promotion do not appear in high-risk branches. This may indicate they are in senior roles where promotion is no longer relevant, such as senior managers and the executive's board. These employees may prioritise stability or nearing retirement, where they do not prioritise in job satisfaction, which could explain their lower attrition risk despite long periods without advancement.

Group C: OverTime, YearsSinceLastPromotion between 1-4 years, JobInvolvement < 3, Environment Satisfaction = 4

Group C (1% of dataset, Predicted chance of attrition: 57%)

Employees who worked overtime, promoted within 1-4 years, with a high environment satisfaction of 4 and a low job involvement (< 3).
Despite receiving a promotion and having high environment satisfaction, these employees are still likely to leave, this might indicate a misalignment between external rewards and personal fulfilment. The low job involvement suggests that they may not feel motivated or supported in their roles, similar in Group B. They might have a lack of purpose or autonomy linked to their work itself, which does not align with ***agentic work principle***, which refers to the extent to which employees feel engaged or empowered in work. Overtime, this mismatch between recognition and actual engagement can lead to a build-up in frustration and hence a desire to leave the company.

Assumptions made

-**Employees behave rationally and uniformly**

It is assumed that employees respond to factors in consistent ways, without accounting for personal preference.

-Past patterns predict future behaviour

It is assumed that the historical attrition behaviours based on the selected variables will propagate the same pattern in the future.

-Promotions are perceived positively and overtime is perceived negatively.

It is assumed that promotions will increase satisfaction and employees do not voluntarily work overtime, hence decreasing satisfaction.

ACTIONABLE RECOMMENDATIONS

Based on the results of both our logistics regression and classification tree models, two key drivers of attrition stood out the most: **Overtime** and **Moderate-Low Job Involvement**. The logistics regression model identified these variables as statistically significant predictors of employee turnover; the classification tree allows us to identify vulnerable employee groups based on specific conditions of the factors. Drawing from both models, we propose the following recommendations.

Recommendation 1: Manage and Reduce Overtime

Firstly, to conduct a **workload audit** across departments to identify the patterns of excessive overtime. Identify employees that are working more than 10 hours per week, the standard working hours in a week for employee is 38 hours in a week (Fair work 2025).
Set **reasonable workload goals**- Such as setting workload caps for junior and mid-level staff, with additional hours requiring formal approval from upper management. In addition, **redistributing tasks load or adjusting team structures** to ensure workloads remain manageable.

As a mitigation measure, **alternative compensation** with overtime- Overtime's compensation is usually monetary, however organisations could also provide time off in lieu or flexible scheduling option to increase employee's autonomy, providing them with more flexibility in their work (Omari 2025). While this does not solve overtime issues directly, it helps reduce its negative impact on employee well-being.

Recommendation 2: Strengthen Job Involvement

Low Job involvement (moderate to low) emerged as a consistent factor that appeared across vulnerable groups. Despite promotions and high environment satisfaction, disengaged employees were still likely to leave. This indicates that external rewards or a supportive working environment are insufficient without meaningful engagement in day-to-day work which strongly driven by intrinsic rewards. (Thomas 2009)

Job involvement increases when employees feel they are learning and progressing in ways that match their interest and roles. (Reference). At Globex Pharma, many roles such as Research and Development, regulatory affairs, or production can become repetitive without ongoing development. This can reduce motivation and emotional connect to their work. Globex can provide **role-specific development opportunities** by offering **skill-enhancing training** that helps employee to excel in their current or desired roles. These learning

opportunities help employees feel more capable in their roles by improving their skills and more likely to found their job meaningful. Eventually, employees **feel seen, valued, and more committed to their everyday work** as company is investing in them. Ultimately, this not merely reduces attrition but also enhances Globex's overall **innovation** and **operational efficiency** as engaged employees are likely to contribute ideas, take initiative and improve their process within the team. (Mahamuda et al. 2011)

DESIGN AND AGILE THINKING

In developing our predictive attrition model for Globex Pharma, we applied design thinking and agile methodologies (Doorely et al. 2018). to ensure the project was feasible and adaptable.

1. Empathize

To address attrition at Globex, we began by understanding the concerns of key stakeholders- HR and Senior management, particularly rising turnover and declining morale. We examined both employee data and external HR research, identifying potential key variables such as Overtime, Environment Satisfaction, Job Involvement and Years since last promotion as consistent factors of attrition.

2. Define

We framed our core issue as: How might we identify key factors of employee attrition to recommend effective retention actions for Globex. Our focus was on identifying predictors, building accurate and effective models and providing practical insights and solutions for vulnerable employee groups.

3. Ideate

We explored multiple model variable selection, including best subset selection and logistics regression, but ultimately prioritised relevance and interpretability by selecting four key predictors. The predictors were not in the best subset selection however we chose it due to its statistical significance ($p<0.05$) to attrition.

4. Prototype

We built two models, logistics regression and classification tree, using four statistically and contextually relevant variables. After running the codes in R studio with the default threshold of 0.5, the sensitivity was extremely low. We then adjusted the threshold to 0.16 to match attrition rate in the dataset, which improved sensitivity and demonstrated agile thinking.

5. Test

Although the revised model produced more false positives, this can be beneficial to Globex by allowing for early intervention. It can then identify and address reasons for rising potential attrition rate before they escalate, supporting stronger retention.

ETHICAL ISSUES

Ethical Concerns Foreseen:

1. Employee Data Confidentiality:

*Fears that sensitive employee information was provided to the model, violating data privacy laws like **The Privacy Act 1988** which protects individual's data collection, use, and dissemination in the public and private sector (NSWGOV 1988.)*

2. Data Ownership and Purpose

Data collected for operational purposes (e.g. payroll, HR records) is now being repurposed for predictive modelling. This raises ethical concerns about whether employees would have agreed to this new usage if asked.

3.Over-reliance on Algorithms/Models

Concerns that the model will be over-utilized and relied upon by Globex Pharma decision-makers, potentially leading to inaccurate information and hit-or-miss insights.

How We Addressed These Issues:

Anonymity: *The Big Data we received (employees2.csv) had no personal identifiers that could be traced back to any individual employees' Identity, maintaining overall **privacy** in all stages of the investigation. Furthermore, in the instance that we were to use any personal data, we implement access control for data analysts to manage and monitor access to sensitive datasets based on role relevance.*

Variable Selection Process: **FOUR** *statistically and business-relevant variables were chosen for the models; the relevance of these models are judged with criteria such as the p-value being less than 0.05. This ensures the model produces outputs reflecting the realistic business environment of Globex Pharma and **lessen the room for potentially biased outputs** by excluding sensitive fields like Age and Gender.*

Human-led Decision Making: *This Report was made with the intention of providing Analytical Models to be used as tools for better informed decision-making instead of replacements for Human Judgement and Real-world context.*

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APPENDIX

VARIABLE SELECTION

```
> library(caret)

> set.seed(123)
> train_indices <- sample(1:nrow(employees2), 0.8 * nrow(employees2))
> train_data <- employees2[train_indices, ]
> test_data <- employees2[-train_indices, ]

> model_all <- glm(Attrition ~ .,data = train_data, family = binomial)
> summary(model_all)
```

LOGISTIC REGRESSION MODEL

```
> train_data$Attrition <- as.factor(train_data$Attrition)
> test_data$Attrition <- as.factor(test_data$Attrition)

> model<- glm(Attrition ~ OverTime + EnvironmentSatisfaction + JobInvolvement +
YearsSinceLastPromotion,
+           data = train, family = "binomial")
> summary(model)

Call:
glm(formula = Attrition ~ OverTime + EnvironmentSatisfaction +
   JobInvolvement + YearsSinceLastPromotion, family = "binomial",
   data = train)

Coefficients:
                    Estimate Std. Error z value Pr(>|z|)
(Intercept)          -0.02927    0.44355  -0.066 0.947379
OverTimeYes           1.49072    0.20232   7.368 1.73e-13 ***
EnvironmentSatisfaction -0.33490    0.08906  -3.760 0.000170 ***
JobInvolvement        -0.47263    0.13729  -3.443 0.000576 ***
YearsSinceLastPromotion  0.01121    0.03057   0.367 0.713724
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 729.42  on 799  degrees of freedom
Residual deviance: 652.43  on 795  degrees of freedom
AIC: 662.43

Number of Fisher Scoring iterations: 5

> # Predict probabilities
> pred1 <- predict(model, newdata = test_data, type = "response")
```

```
>
>
> # Use threshold = 0.16 instead of 0.5
> class1 <- ifelse(pred1 > 0.16, "Yes", "No")
> class1 <- factor(class1, levels = c("Yes", "No"))
>
> # Ensure levels match
> test$Attrition <- factor(test$Attrition, levels = c("Yes", "No"))
>
> # Evaluate performance
> confusionMatrix(class1, test$Attrition, positive = "Yes")
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  116  11
       Yes  56  17

               Accuracy : 0.665
                 95% CI : (0.595, 0.73)
    No Information Rate : 0.86
    P-Value [Acc > NIR] : 1

                  Kappa : 0.1683

 Mcnemar's Test P-Value : 7.639e-08

            Sensitivity : 0.6071
            Specificity : 0.6744
         Pos Pred Value : 0.2329
         Neg Pred Value : 0.9134
             Prevalence : 0.1400
         Detection Rate : 0.0850
   Detection Prevalence : 0.3650
      Balanced Accuracy : 0.6408

       'Positive' Class : Yes
```

CLASSIFICATION TREE

```
library(rpart.plot)

> tree_model <- rpart(Attrition ~ OverTime + EnvironmentSatisfaction + JobInvolvement +
YearsSinceLastPromotion, data = train_data,  method = "class", control = rpart.control(cp = 0.001))
> rpart.plot(tree_model)
> summary(tree_model)

Call:
rpart(formula = Attrition ~ OverTime + EnvironmentSatisfaction +
    JobInvolvement + YearsSinceLastPromotion, data = train_data,
    method = "class", control = rpart.control(cp = 0.001))
  n= 800

          CP nsplit rel error   xerror      xstd
```

```
1 0.007692308     0 1.0000000 1.000000 0.08026399
2 0.002564103     6 0.9538462 1.115385 0.08381395
3 0.001000000     9 0.9461538 1.115385 0.08381395

Variable importance
          OverTime YearsSinceLastPromotion EnvironmentSatisfaction        JobInvolvement
                54                    19                    16                    11

Node number 1: 800 observations,    complexity param=0.007692308
  predicted class=No   expected loss=0.1625  P(node) =1
    class counts:   670   130
   probabilities: 0.838 0.163
  left son=2 (582 obs) right son=3 (218 obs)
  Primary splits:
      OverTime             splits as  LR,      improve=14.2158400, (0 missing)
      EnvironmentSatisfaction < 1.5  to the right, improve= 2.5630330, (0 missing)
      JobInvolvement        < 1.5  to the right, improve= 1.8301330, (0 missing)
      YearsSinceLastPromotion < 14.5 to the left,  improve= 0.2144208, (0 missing)

Node number 2: 582 observations
  predicted class=No   expected loss=0.104811  P(node) =0.7275
    class counts:   521    61
   probabilities: 0.895 0.105

Node number 3: 218 observations,    complexity param=0.007692308
  predicted class=No   expected loss=0.3165138  P(node) =0.2725
    class counts:   149    69
   probabilities: 0.683 0.317
  left son=6 (183 obs) right son=7 (35 obs)
  Primary splits:
      EnvironmentSatisfaction < 1.5  to the right, improve=3.261616, (0 missing)
      JobInvolvement        < 3.5  to the right, improve=1.045739, (0 missing)
      YearsSinceLastPromotion < 4.5  to the left,  improve=0.440568, (0 missing)

Node number 6: 183 observations,    complexity param=0.007692308
  predicted class=No   expected loss=0.2786885  P(node) =0.22875
    class counts:   132    51
   probabilities: 0.721 0.279
  left son=12 (17 obs) right son=13 (166 obs)
  Primary splits:
      JobInvolvement        < 3.5  to the right, improve=1.8118990, (0 missing)
      YearsSinceLastPromotion < 4.5  to the left,  improve=0.8974973, (0 missing)
      EnvironmentSatisfaction < 2.5  to the right, improve=0.3563792, (0 missing)

Node number 7: 35 observations,    complexity param=0.007692308
  predicted class=Yes  expected loss=0.4857143  P(node) =0.04375
    class counts:    17    18
   probabilities: 0.486 0.514
  left son=14 (8 obs) right son=15 (27 obs)
  Primary splits:
      YearsSinceLastPromotion < 2.5  to the right, improve=0.40238100, (0 missing)
      JobInvolvement        < 2.5  to the right, improve=0.04126984, (0 missing)

Node number 12: 17 observations
  predicted class=No   expected loss=0.05882353  P(node) =0.02125
    class counts:    16     1
   probabilities: 0.941 0.059
```

```
Node number 13: 166 observations,    complexity param=0.007692308
  predicted class=No   expected loss=0.3012048  P(node) =0.2075
    class counts:   116    50
   probabilities: 0.699 0.301
  left son=26 (14 obs) right son=27 (152 obs)
  Primary splits:
      YearsSinceLastPromotion < 7.5  to the right, improve=0.7667361, (0 missing)
      EnvironmentSatisfaction < 2.5  to the right, improve=0.5739625, (0 missing)
      JobInvolvement          < 2.5  to the right, improve=0.2908211, (0 missing)

Node number 14: 8 observations
  predicted class=No   expected loss=0.375  P(node) =0.01
    class counts:     5     3
   probabilities: 0.625 0.375

Node number 15: 27 observations
  predicted class=Yes  expected loss=0.4444444  P(node) =0.03375
    class counts:    12    15
   probabilities: 0.444 0.556

Node number 26: 14 observations
  predicted class=No   expected loss=0.1428571  P(node) =0.0175
    class counts:    12     2
   probabilities: 0.857 0.143

Node number 27: 152 observations,    complexity param=0.007692308
  predicted class=No   expected loss=0.3157895  P(node) =0.19
    class counts:   104    48
   probabilities: 0.684 0.316
  left son=54 (133 obs) right son=55 (19 obs)
  Primary splits:
      YearsSinceLastPromotion < 4.5  to the left,  improve=3.0075190, (0 missing)
      EnvironmentSatisfaction < 2.5  to the right, improve=0.4970083, (0 missing)
      JobInvolvement          < 2.5  to the right, improve=0.4912718, (0 missing)

Node number 54: 133 observations,    complexity param=0.002564103
  predicted class=No   expected loss=0.2781955  P(node) =0.16625
    class counts:    96    37
   probabilities: 0.722 0.278
  left son=108 (71 obs) right son=109 (62 obs)
  Primary splits:
      YearsSinceLastPromotion < 0.5  to the right, improve=0.8506079, (0 missing)
      JobInvolvement          < 2.5  to the right, improve=0.7201350, (0 missing)
      EnvironmentSatisfaction < 2.5  to the right, improve=0.5103847, (0 missing)
  Surrogate splits:
      EnvironmentSatisfaction < 3.5  to the left,  agree=0.586, adj=0.113, (0 split)

Node number 55: 19 observations
  predicted class=Yes  expected loss=0.4210526  P(node) =0.02375
    class counts:     8    11
   probabilities: 0.421 0.579

Node number 108: 71 observations,    complexity param=0.002564103
  predicted class=No   expected loss=0.2253521  P(node) =0.08875
    class counts:    55    16
   probabilities: 0.775 0.225
```

```
  left son=216 (49 obs) right son=217 (22 obs)
  Primary splits:
     JobInvolvement        < 2.5  to the right, improve=1.2191590, (0 missing)
     YearsSinceLastPromotion < 3.5  to the right, improve=0.4705506, (0 missing)
     EnvironmentSatisfaction < 3.5  to the left,  improve=0.2173038, (0 missing)

Node number 109: 62 observations
  predicted class=No   expected loss=0.3387097  P(node) =0.0775
    class counts:    41    21
   probabilities: 0.661 0.339

Node number 216: 49 observations
  predicted class=No   expected loss=0.1632653  P(node) =0.06125
    class counts:    41     8
   probabilities: 0.837 0.163

Node number 217: 22 observations,    complexity param=0.002564103
  predicted class=No   expected loss=0.3636364  P(node) =0.0275
    class counts:    14     8
   probabilities: 0.636 0.364
  left son=434 (15 obs) right son=435 (7 obs)
  Primary splits:
     EnvironmentSatisfaction < 3.5  to the left,  improve=0.8865801, (0 missing)

Node number 434: 15 observations
  predicted class=No   expected loss=0.2666667  P(node) =0.01875
    class counts:    11     4
   probabilities: 0.733 0.267

Node number 435: 7 observations
  predicted class=Yes  expected loss=0.4285714  P(node) =0.00875
    class counts:     3     4
   probabilities: 0.429 0.571

> conf_matrix <- confusionMatrix(factor(tree_predictions),factor(test_data$Attrition), positive = "Yes")
> print(conf_matrix)
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  162  28
       Yes   4   6

               Accuracy : 0.84
                 95% CI : (0.7817, 0.8879)
    No Information Rate : 0.83
    P-Value [Acc > NIR] : 0.3963

                  Kappa : 0.2118

 Mcnemar's Test P-Value : 4.785e-05

            Sensitivity : 0.1765
            Specificity : 0.9759
         Pos Pred Value : 0.6000
         Neg Pred Value : 0.8526
             Prevalence : 0.1700
```

```
Detection Rate : 0.0300
Detection Prevalence : 0.0500
Balanced Accuracy : 0.5762

'Positive' Class : Yes
```